

ADITYA

Aerodynamics Engineer | CFD & Aerodynamic Design

London, UK | +44 7549 349926 | aditya.pravega@gmail.com | [linkedin.com/in/aditya-kumar-78010b185](https://www.linkedin.com/in/aditya-kumar-78010b185)

PROFILE

Aeronautical engineer finishing an MSc at Imperial College London, with three years of hands-on aerodynamic design and CFD, most of it on Formula Student cars. I have owned a full aero package more than once: front wing, rear wing, diffuser and undertray, from concept CAD in CATIA V5 through development in STAR-CCM+ to parts that were made and raced. Comfortable taking a case end to end, from geometry clean-up and meshing through solver setup, post-processing and validation against test data, and used to defending the result in front of people who question every assumption in it. My MSc thesis sits a level below the tools, deriving the RANS equations in a rotating frame and adapting the turbulence models to match. Looking for an entry-level aerodynamics role where the loop between a design decision and its effect on the car is short.

TECHNICAL SKILLS

Aerodynamic Design: Front and rear wing design, diffuser and undertray development, cooling duct and bodywork design; aerofoil selection and planform iteration; parametric design sweeps and optimisation; generative design; aerodynamic and vehicle dynamics trade studies

CFD & Simulation: STAR-CCM+, Ansys Fluent, OpenFOAM, XFLR5, PyEuler2D (Python Euler solver); RANS turbulence modelling (Spalart-Allmaras, $k-\omega$ / $k-\epsilon$), Moving Reference Frame (MRF), multiphase VOF modelling, conjugate heat transfer and thermal analysis, mesh generation and mesh independence studies, post-processing and validation against test data

CAD & Geometry: CATIA V5, Siemens NX, SolidWorks, Fusion 360, AutoCAD; parametric modelling, geometry clean-up and simulation preparation

Programming & Automation: Python (solver development, post-processing, workflow automation), STAR-CCM+ Java macro automation, MATLAB, HPC and cluster job submission, Linux command line

Vehicle Performance: Lap-time simulation (QSLS), Pacejka 6.1 tyre modelling, vehicle dynamics integration, structural and finite element analysis

EDUCATION

Imperial College London

MSc Advanced Aeronautical Engineering

Sep 2025 – Present

London, UK

- Modules: Turbulence Modelling, Applied Aerodynamics, Flow Instability and Transition, Design Optimisation, Finite Elements, Lightweight Structures.

Vellore Institute of Technology

B.Tech Mechanical Engineering (Automotive Specialisation), CGPA 8.66 / 10

Sep 2021 – Jul 2025

Vellore, India

- Modules: Computational Fluid Dynamics, Fluid Dynamics, Vehicle Body Aerodynamics, Thermal Engineering, Thermodynamics, Vehicle Dynamics, Solid Mechanics, Control Systems.

MOTORSPORT AERODYNAMICS

Imperial Formula Racing

Simulations Engineer

Oct 2025 – Present

London, UK

- Designing the IFR-27 front and rear wings in CATIA V5 and developing them in STAR-CCM+, chasing downforce without giving away aerodynamic efficiency.
- Integrated the Pacejka 6.1 tyre model into the team's QSLs lap-time simulation, so an aero change can be judged in lap time rather than in coefficients on their own.
- Carrying the aerodynamics and vehicle dynamics workstreams at the same time, against a season calendar that does not move.

Pravega Racing | Formula Student, SAE, VIT

CFD Head

Dec 2023 – Jan 2025

Vellore, India

- Owned the whole aerodynamic package — front wing, rear wing, diffuser, undertray — designed in CATIA V5 and developed in STAR-CCM+.
- Ran the CFD and aerodynamic validation for both the electric and the combustion cars: two sets of downforce, structure and weight trade-offs on one calendar.
- Took the thermal CFD on the battery pack, motor and motor control unit, confirming they stayed inside their limits under race conditions.
- Wrote the simulation workflows down so someone else could repeat them, and taught newer members meshing practice and the theory underneath it.

- The team finished 1st in both the Cost and Engineering Design events and 3rd overall at Formula Bharat 2025.

Pravega Racing | Formula Student, SAE, VIT

Mar 2022 – Dec 2023

Aerodynamics Engineer

Vellore, India

- Designed the full aero package: wings, cooling ducts and bodywork, following it from concept CAD through CFD to parts that were actually made.
- Ran the aero and vehicle dynamics numbers together to work out what the aerodynamics was really worth in braking, acceleration and cornering.
- Wrote and presented the technical design reports and cost analyses to judging panels at Formula Student Germany and Formula Student East, and answered for every decision in them.

RESEARCH

Imperial College London, with ESDU (Accuris)

Feb 2026 – Present

MSc Thesis

London, UK

“Derivation of the RANS Equations for Rotational Flow via Transformation Between Axes of Translation and Rotation”

- Derived the transformation of the 2D RANS equations from a stationary frame into a Moving Reference Frame (MRF), term by term.
- Rewrote the Spalart-Allmaras and k- ω turbulence models to sit inside that formulation, so they hold up in the unsteady flow around a rotating propeller.
- Validating rotating NACA 0012 cases in Ansys Fluent against PyEuler2D, a Python Euler solver built for the project, and tracking down why they diverge: domain sizing, rotating-frame setup, and how far the far-field boundary has to sit.

Vellore Institute of Technology

Sep 2024 – Nov 2024

Undergraduate Final Year Project

Vellore, India

“Design and Development of an FSAE Rear Wing Mount using Generative Design and CFD”

- Used generative design to take 50% of the mass out of the rear wing mount without losing structural performance.
- Put the resulting geometry back through CFD, which came out 24% better on aerodynamic efficiency.

ADDITIONAL ENGINEERING EXPERIENCE

Adani Defence and Aerospace

Jan 2025 – Apr 2025

Project Trainee

New Delhi, India

- Owned the airframe architecture and aerodynamic design of a loitering munition, from first concept through to validation.
- Iterated wing profiles and planforms until lift-to-drag, endurance and stability came out together rather than being traded against one another.
- Sized the control surfaces and ran the stability analysis, then worked the numbers back into something that could actually be built.
- Selected the lightweight composite structures, balancing strength-to-weight against the aerodynamic shape they had to hold.

Tata Motors Passenger Vehicles Limited

Sep 2023 – Oct 2023

Project Trainee

Ahmedabad, India

- Analysed logged vehicle sensor data to see how sensor behaviour fed through into performance and efficiency.
- Ran an FMEA across the sensor subsystems, ranking failure modes by risk so the weakest points in the integration got attention first.

American Society of Mechanical Engineers (VIT Chapter)

Dec 2023 – Jan 2025

Event Coordinator

Vellore, India

- Ran the chapter's technical seminars and workshops on CFD and aerodynamics, and coached juniors through their first CFD work.

ACHIEVEMENTS

- Formula Bharat 2025: 1st in Cost, 1st in Engineering Design, 1st overall in Statics, 3rd overall (Combustion).
- Formula Student Germany 2023: 18th of 160 teams worldwide, 9th in Statics (Hockenheimring, Germany).
- Formula Student East 2023: 12th of 150 teams worldwide (Hungaroring, Hungary).
- Formula Bharat 2023: 2nd in Statics, 7th overall nationally (Kari Motor Speedway, Coimbatore).
- Vortex 360 Designathon by Autodesk, Feb 2024: 1st of 200 competing teams in the Design Challenge.
- VIT University Day 2023: Achiever and Special Achiever awards for national and international competition results; formally recognised by SAE-VIT for contributions as CFD Head of Pravega Racing.